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1.A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

A) Introduction

Remote sensing satellites capture images of the same geographical area at different times. These images help scientists monitor environmental changes such as:

- Deforestation
- Floods
- Urban development
- Crop growth
- Coastal changes
- Glacier melting

Before comparing these images, they must be perfectly aligned. This process is called Image Registration.

Initially, the system uses Edge Detection to find matching regions. However, because satellite images are captured under different conditions, the alignment becomes inaccurate.To solve this problem, the development team replaces edge detection with Harris Corner Detection, which provides more reliable feature points.

Problem Statement

The existing image registration system depends on edge detection for finding matching regions. The system faces several problems because:

- Images are captured at different times.
- Lighting conditions change.
- Satellite viewing angle changes.
- Background objects change.
- Seasonal variations affect the appearance.

As a result,

- Wrong regions are matched.
- Images are not aligned correctly.
- Environmental monitoring becomes inaccurate.

Limitations of Edge Detection

Edge detection identifies only the boundaries of objects.
Although edges are useful, they are not always reliable for satellite image matching.

1. Sensitive to Illumination Changes

Different sunlight conditions create different edge strengths.

Example: Morning image , Evening image

The same building may produce different edges.

Result:

Incorrect matching.

2. Sensitive to Noise

Satellite images often contain

- Sensor noise
- Atmospheric disturbances
- Cloud shadows

Edge detectors detect unwanted edges created by noise.

Result:

Many false edges appear.

3. Difficult to Identify Unique Locations

Edges extend over long distances.
Many pixels along the same edge look identical.

Example:

Road boundaries

The algorithm cannot determine the exact matching point.

Result:

Poor localization.

4. Rotation Problems

If the satellite captures the area from a different angle,
Edges appear rotated.
Traditional edge matching cannot easily recognize rotated edges.

Result:

Alignment accuracy decreases.

5. Background Changes

Environmental changes like

- New buildings
- Trees
- Rivers
- Vehicles

create new edges.
Old edges disappear.

Result:

Edge matching becomes unreliable.

6. Weak Repeatability

The same edges are not always detected in every image.
Because of

- Weather
- Lighting
- Resolution

edge locations change.
Result:
Image registration fails.

Why Harris Corner Detection is Better
Harris Corner Detection identifies corner points instead of long edges.
Corners are locations where the image intensity changes in multiple directions.
Examples:

- Building corners
- Road intersections
- Bridge corners
- Field boundaries
- Roof corners

These locations remain stable even when image conditions change.

Working Principle of Harris Corner Detection
The Harris algorithm works in four main steps.
Step 1: Calculate Image Gradients
The image intensity changes are calculated in

- Horizontal direction (X)
- Vertical direction (Y)

This helps identify areas with large intensity variation.

Step 2: Build the Structure Matrix
The algorithm studies how pixel intensity changes inside a small window.
This matrix contains information about

- Horizontal changes
- Vertical changes
- Combined changes

Step 3: Compute Harris Response
The Harris response value is calculated.
High response value
→ Corner
Medium response
→ Edge
Low response
→ Flat region
Step 4: Select Strong Corners
Only the strongest corners are selected.

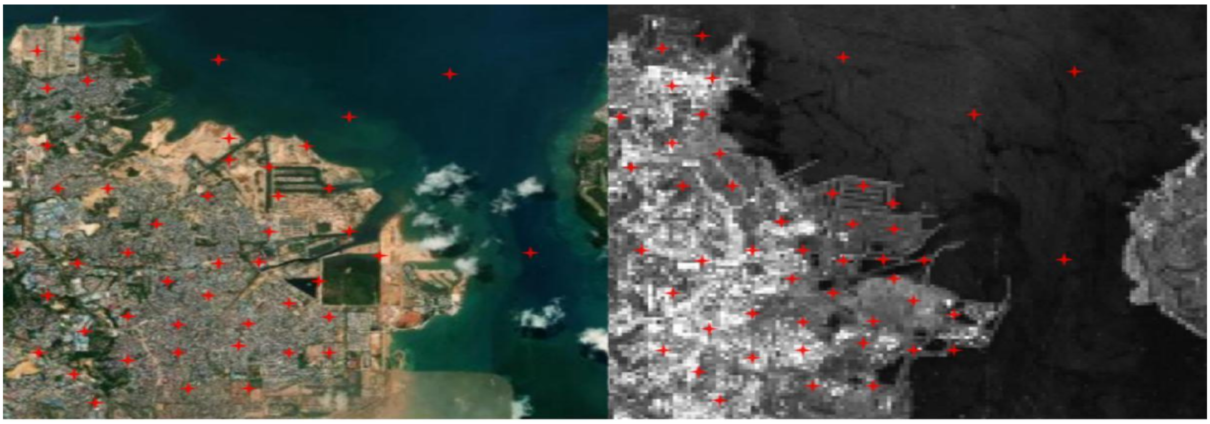
These points become feature points for image matching.

Applications

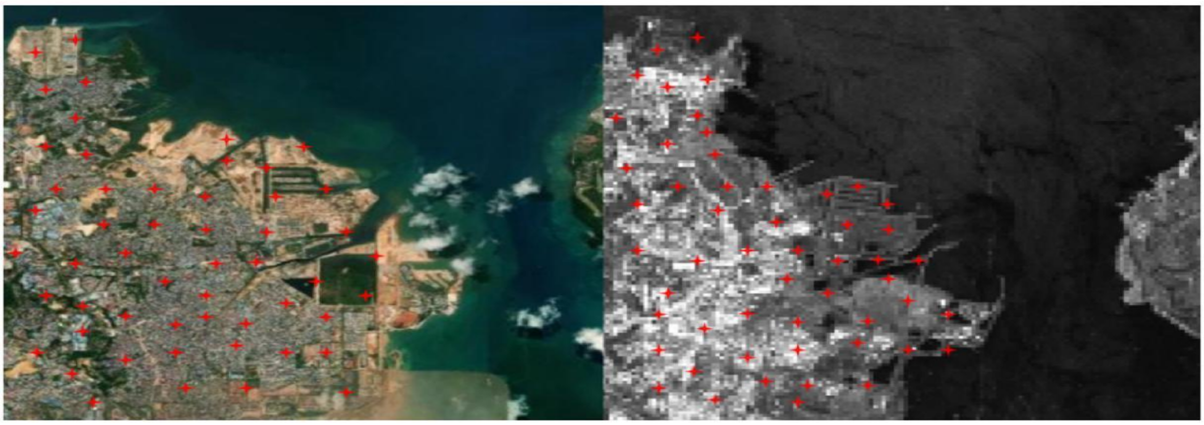
- Forest monitoring
- Flood detection
- Disaster assessment
- Land-use analysis
- Urban planning
- Agricultural monitoring
- Coastal erosion monitoring

Conclusion

Edge detection is useful for identifying object boundaries but is not reliable for satellite image registration because it is sensitive to lighting, noise, rotation, and background changes. Harris Corner Detection identifies stable and unique corner points that remain consistent under different imaging conditions. This improves feature localization, matching accuracy, repeatability, and overall robustness. Therefore, Harris Corner Detection is a better choice for accurate satellite image registration and long-term environmental monitoring.



(a) Feature points detection by multi-scale Harris



(b) Feature points detection by PC-Harris

2.An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

A) Introduction

An autonomous surveillance system continuously captures high-resolution images to detect

- Vehicles
- Pedestrians
- Cyclists
- Other moving objects

The camera captures color images (RGB).

Processing these large color images directly requires high computational power, making the system slower.To improve speed, engineers consider converting images into

- Grayscale
- Binary

before applying feature detection algorithms.The goal is to determine which image representation provides the best balance between speed and detection accuracy.

Problem Statement

The surveillance system must detect objects in real time.

Problems with color images include:

- High memory usage
- Slow processing
- Increased computational cost
- Delayed object detection

The engineers analyze different image representations to improve performance.

Types of Image Representation

1. Color Image (RGB)

A color image contains three channels:

- Red
- Green
- Blue

Each pixel stores three intensity values.

Advantages

- Rich visual information
- Useful for object recognition
- Better appearance

Disadvantages

- Large memory requirement

- Slow processing
- High computational complexity

2. Grayscale Image
A grayscale image contains only one intensity value per pixel.
Brightness ranges from
0 (black)
to
255 (white)
Advantages

- Smaller memory
- Faster processing
- Preserves important shapes
- Good for edge detection
- Suitable for feature extraction

Disadvantages

- Color information is lost.

3. Binary Image
Binary images contain only two values:

- Black
- White

Pixels are obtained using thresholding.
Advantages

- Very fast
- Lowest memory requirement

Disadvantages

- Many image details disappear.
- Sensitive to threshold selection.
- Difficult to detect objects in complex scenes.

Impact on Computational Efficiency
Color Images
Processing three channels requires

- More calculations
- More memory
- More processing time

This is why many computer vision algorithms first convert images to grayscale.

Real-Time Detection Process

1. Capture RGB image.
2. Convert RGB image to grayscale.
3. Apply noise reduction.

4. Detect edges.
5. Extract features.
6. Detect vehicles and pedestrians.
7. Draw bounding boxes.
8. Display results in real time.

Applications

- Traffic monitoring
- Smart cities
- Autonomous vehicles
- Security surveillance
- Parking management
- Pedestrian detection
- Road safety systems

Conclusion

Color images contain rich information but require high computational power, making them less suitable for real-time processing. Binary images are extremely fast but lose important image details, reducing detection accuracy. Grayscale images provide the best balance between speed, memory usage, edge detection quality, and feature extraction. Therefore, grayscale representation is the most appropriate choice for real-time surveillance systems that need accurate and reliable vehicle and pedestrian detection.